Examiner only

8. *(a)* (i) Define the term *standard enthalpy change of combustion*, $\Delta_c H^\theta$. [2]

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(ii) The standard enthalpy change of combustion of ethane is –1561 kJ mol^{-1}.

The values of some average bond enthalpies are shown in the table below.

Bond	Average bond enthalpy / kJ mol^{-1}
C—C	348
O=O	495
C=O	799
O—H	463

$$C_2H_6 \quad + \quad \ldots\ldots\ldots \; O_2 \longrightarrow 2CO_2 \quad + \quad 3H_2O$$

Balance the equation for the combustion of ethane and use the information given to calculate the average bond enthalpy for a C—H bond. [4]

Bond enthalpy = ... kJ mol^{-1}

(iii) Suggest a reason why bond enthalpies are described as being **average** bond enthalpies. [1]

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(b) Charcoal consists mainly of carbon. It has been produced for many centuries by heating wood in the absence of oxygen. Natural gas consists mainly of methane and is obtained from underground sources. It was formed in a similar way to coal and oil.

The enthalpy changes of combustion of carbon and methane are in the table.

Substance	Enthalpy change of combustion / kJ mol^{-1}
carbon, C	–394
methane, CH_4	–889

Two students were discussing the use of charcoal and methane as fuels.

One said that 'methane produced more heat per gram when burned so that it was a better fuel'.

The other student said that 'the use of charcoal contributed less to the overall increase of carbon dioxide levels in the atmosphere'.

Discussion these two statements. [6 QER]

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8. *(a)* Compound **X** is a hydrocarbon that contains 85.6% of carbon by mass. The relative molecular mass of **X** is in the range 50-60.

11.2g of compound **X** decolourised exactly 32.0g of bromine in the absence of light. Further bromine was decolourised when the reaction mixture was placed in direct sunlight.

(i) Find the empirical formula of **X** and hence its molecular formula. Show clearly how you carried out your calculation. [3]

Molecular formula ....

(ii) Explain what can be deduced from the fact that 11.2g of **X** reacts with exactly 32.0g of bromine. [2]

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(iii) What type of reaction mechanism occurs when **X** reacts with bromine in the absence of light? [1]

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(iv) What type of reaction mechanism occurs when **X** reacts with more bromine in the presence of sunlight? [1]

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(v) Draw displayed formulae for each of the following. [3]

Compound **X**

The product formed when **X** reacts with bromine in the absence of light

One possible product formed when **X** reacts with excess bromine in the presence of sunlight

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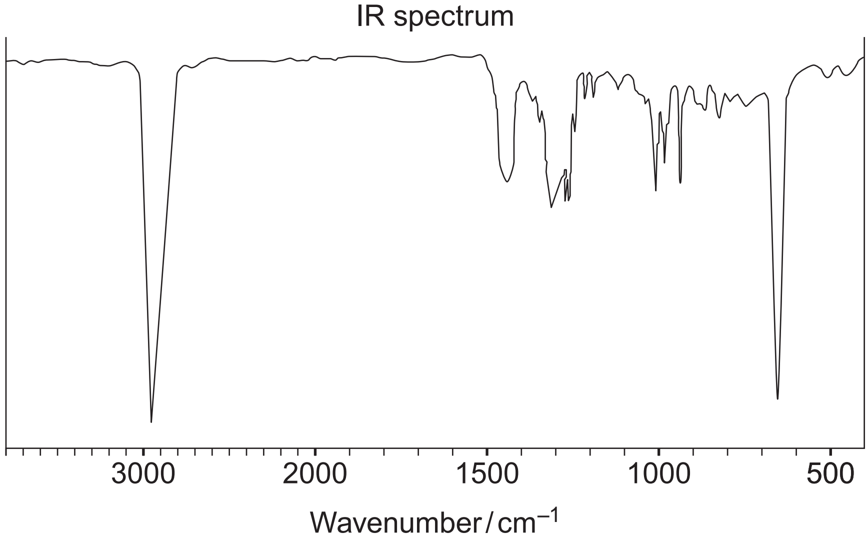
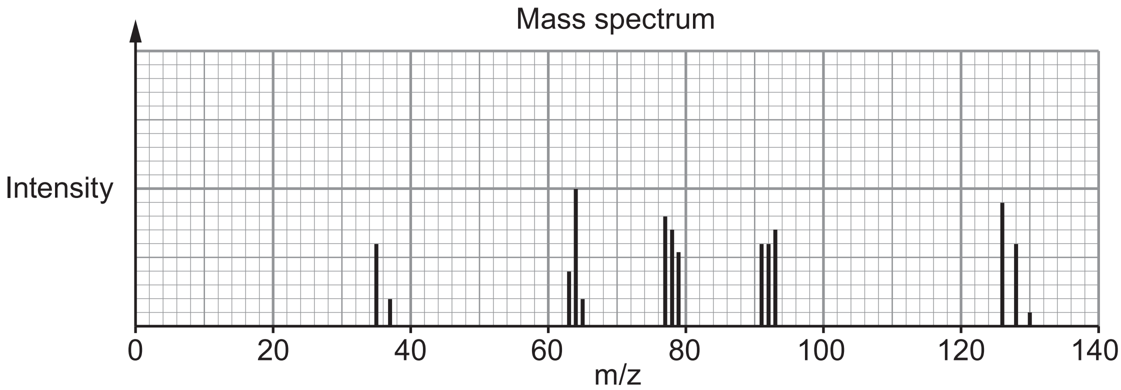
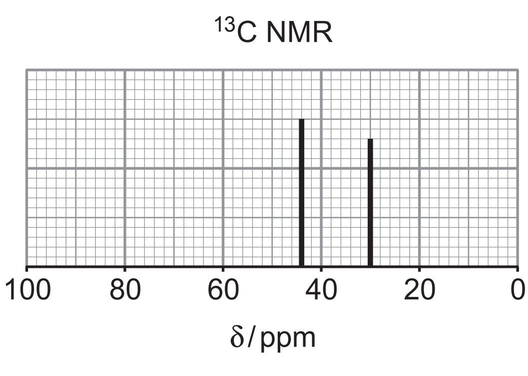
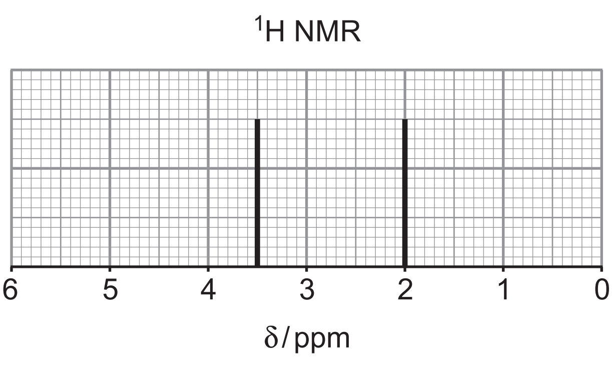
(b) Compound **Y** is a halogenoalkane. A simplified form of the IR spectrum, the mass spectrum, the ^{13}C NMR spectrum and the low resolution ^{1}H NMR spectrum of **Y** are shown below.









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Use these spectra to determine the identity of compound **Y**. You must use evidence from each spectrum and explain how you used it. [6 QER]

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SECTION B

*Answer **all** questions in the spaces provided.*

8. A student was told that he could prepare chloroethane, C_2H_5Cl, by mixing ethane with chlorine. He added 2.0 g of ethane to excess chlorine and left the mixture exposed to ultraviolet light for several hours. He was then able to use a university laboratory to see whether chloroethane had been made.

(a) State an instrumental method by which the sample could be analysed. Explain how this would show that chlorination had occurred. [2]

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(b) State why it is necessary to use ultraviolet light when making chloroethane from ethane. Give equations to show the mechanism for the formation of chloroethane. [4]

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(c) The student found that he had made 1.0 g of chloroethane. Calculate his percentage yield. [3]

Percentage yield = ... %

(d) The student was disappointed by the yield he obtained but his teacher told him that the yield was always poor due to other products being formed in this reaction.

Identify **two** other organic products, apart from chloroethane, and explain how they are formed. [2]

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8. Many factors affect the rate of a chemical reaction.

(a) Explain why a change in concentration affects the rate of a reaction. [2]

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(b) Under certain conditions nitrogen monoxide reacts with carbon monoxide and oxygen according to the equation below.

$$NO(g) + CO(g) + O_2(g) \longrightarrow NO_2(g) + CO_2(g)$$

A study of the rate of this reaction, using varying initial concentrations of nitrogen monoxide and oxygen, gave the following data.

Experiment number	Concentration NO / mol dm^{-3}	Concentration O_2 / mol dm^{-3}	Initial rate of reaction / mol dm^{-3}s^{-1} × 10^{-4}
1	1.0×10^{-4}	1.0×10^{-4}	4.40
2	2.0×10^{-4}	1.0×10^{-4}	17.6
3	3.0×10^{-4}	1.0×10^{-4}	39.6
4	2.0×10^{-4}	2.0×10^{-4}	17.6

(i) Use the data to determine how the concentration of NO affects the rate of the reaction. Explain your answer by referring to the experiment numbers. [2]

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(ii) Use the data to determine how the concentration of O_2 affects the rate of the reaction. Explain your answer by referring to the experiment numbers. [2]

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(iii) Suggest a method that could be used to follow changes over time as the reaction of nitrogen monoxide, carbon monoxide and oxygen proceeds. Explain your answer. [2]

$$NO(g) + CO(g) + O_2(g) \longrightarrow NO_2(g) + CO_2(g)$$

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(iv) Describe the environmental implications on the atmosphere if the reaction above occurs on a large scale. [2]

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(v) The reaction shown is one of many that are catalysed in a catalytic converter. State where in a car a catalytic converter is used and name a suitable catalyst. [2]

Catalytic converter used in ....

Catalyst used ....

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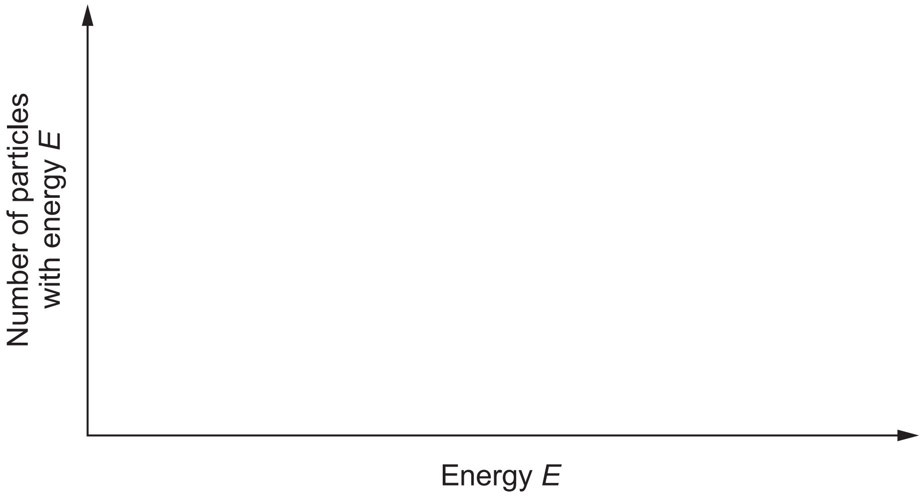
(c) On the axes below draw a Boltzmann distribution curve of the energies of molecules at a certain temperature.

Use this curve to explain how a catalyst increases the rate of a reaction. [3]



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8. Compound **A** contains only carbon, hydrogen and an unknown halogen.

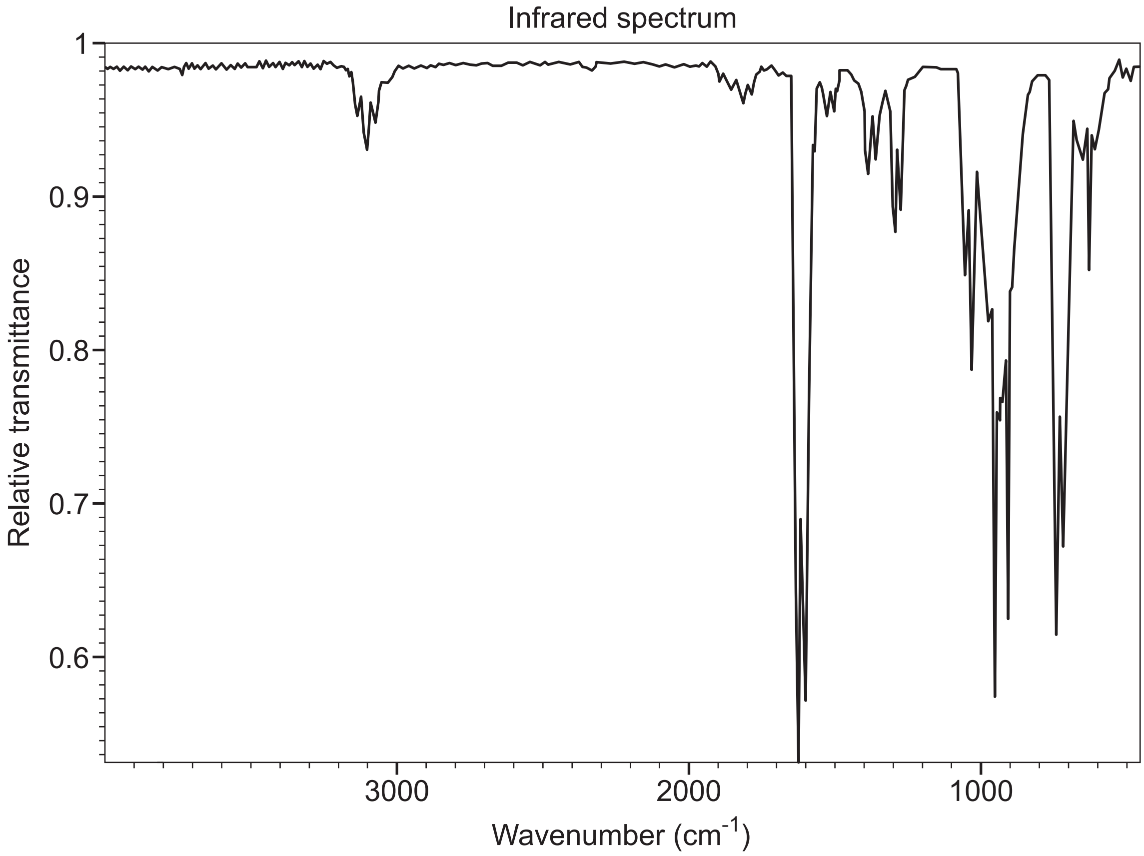
Refluxing compound **A** in aqueous sodium hydroxide followed by the addition of nitric acid and aqueous silver nitrate produces a white precipitate.

Elemental analysis of compound **A** indicates it contains 39.02% carbon and 3.25% hydrogen by mass.

When bromine is added to compound **A**, 123 g of compound **A** reacts with 320 g of bromine.

The ^{1}H NMR spectrum of compound **A** consists of only one peak. The ^{13}C NMR spectrum of compound **A** consists of two peaks.

The infrared spectrum and simplified mass spectrum are shown below and overpage.



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SECTION B

Answer **all** questions.

8. The use of biofuels has been proposed as the way to ensure the sustainability of energy production in the future.

Discuss the extent to which you agree with this proposal, giving both advantages and disadvantages of the use of biofuels. Include relevant examples in your answer. [6 QER]

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